Computational Logic — Exercise Workbook

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How to use this workbook

All exercises from the two official lists (Propositional Logic, Predicate Logic 2026-2027), grouped by chapter and method. Tick each item when you can solve it alone; worked solutions are in the companion doc *Computational Logic — Solutions & Theory*.

Required solving format (from the exercise sheet):

1. Write the problem statement.
2. Present the theory needed: concepts, definitions, theorems, method/algorithm.
3. Apply the method from step 2.
4. State the conclusion based on step 3, in accordance with the theory.

Notation: ¬ not · ∧ and · ∨ or · → implies · ↔ equivalent · ↓ NOR · ↑ NAND · ⊕ XOR · ⊨ logical consequence · ⊢ deduction. Precedence, strongest first: ¬, ∧, ∨, →, ↔.

Legend: (S) = solved in seminar, the model solution is in your notes.

Part I — Propositional Logic

Chapter 1 — Semantics: the truth table method

Exercise 1 — Classify the formula

Using the truth table method decide what kind of formula (consistent, inconsistent, tautology, contingent) is U_j. Write all the models and anti-models of U_j.

- [ ] $U_1 = q \vee \neg p \vee r \to \neg(p \vee r)$
- [ ] $U_2 = \neg p \vee \neg(q \wedge r) \to q \wedge \neg p$
- [ ] $U_3 = \neg p \wedge (\neg q \vee r) \to q \wedge \neg p \vee r$
- [ ] $U_4 = \neg(\neg p \wedge q) \vee r \to p \wedge (\neg q \vee r)$
- [ ] $U_5 = \neg p \vee q \wedge r \to q \wedge \neg p$ **(S)**
- [ ] $U_6 = \neg p \vee (\neg q \wedge \neg r) \to q \wedge \neg p \wedge r$

Exercise 2 — Prove the tautologies

Prove that the following formulas are tautologies using the truth table method.

- [ ] Left-distribution of → over ∧: $(p \to (q \wedge r)) \to ((p \to q) \wedge (p \to r))$

- [ ] Permutation of the premises: $(p \to (q \to r)) \to (q \to (p \to r))$
- [ ] Reunion of the premises: $(p \to (q \to r)) \to (p \land q \to r)$
- [ ] Separation of the premises: $(p \land q \to r) \to (p \to (q \to r))$
- [ ] Left-distribution of $\lor$ over $\to$: $p \lor (q \to r) \to ((p \lor q) \to (p \lor r))$ **(S)**
- [ ] Left-distribution of $\to$ over $\lor$: $(p \to (q \lor r)) \to ((p \to q) \lor (p \to r))$

Exercise 3 — Properties of ↓ (NOR), ↑ (NAND), ⊕ (XOR)

Check the following properties using the truth table method.

- [ ] Associativity of $\uparrow$: $p \uparrow (q \uparrow r) \equiv (p \uparrow q) \uparrow r$
- [ ] Associativity of $\downarrow$: $p \downarrow (q \downarrow r) \equiv (p \downarrow q) \downarrow r$
- [ ] Associativity of $\oplus$: $p \oplus (q \oplus r) \equiv (p \oplus q) \oplus r$ **(S)**
- [ ] Distribution of $\uparrow$ over $\downarrow$: $p \uparrow (q \downarrow r) \equiv (p \uparrow q) \downarrow (p \uparrow r)$
- [ ] Distribution of $\downarrow$ over $\uparrow$: $p \downarrow (q \uparrow r) \equiv (p \downarrow q) \uparrow (p \downarrow r)$ **(S)**
- [ ] De Morgan's laws for $\downarrow$ and $\uparrow$: $\neg(p \downarrow q) \equiv \neg p \uparrow \neg q$ and $\neg(p \uparrow q) \equiv \neg p \downarrow \neg q$

Exercise 4 — Logical consequence

Using the truth table method, check whether the following logical consequences hold or not.

- [ ] $p \to q \models (p \to r) \to (p \to q \land r)$
- [ ] $p \to q \models (q \to r) \to (p \to r)$
- [ ] $p \to (q \to r) \models (p \to q) \to (p \to r)$
- [ ] $p \to r \models (q \to r) \to ((p \lor q) \to r)$
- [ ] $p \to q \models (q \to r) \to (p \to q \lor r)$ **(S)**
- [ ] $r \to (q \to p) \models (r \to q) \to (r \to p)$

Exercise 5 — From natural language to formulas

Translate the sentences S1, S2, S3 into propositional formulas and find their models and anti-models.

- [ ] **Set 1 (S)** — S1: If it is sunny and warm then I will go swimming. S2: Mary and Jane will go for a walk only if it is sunny and not windy. S3: John goes to the cinema if and only if it is Saturday and it is raining.
- [ ] **Set 2** — S1: We will go skating if and only if it is cold outside and the rink is open. S2: Skating is easy if you have experience and you like the cold. S3: You can enter the rink only if you bring skates from home or borrow skates from the rink.

- [ ] **Set 3 (S)** — S1: Sherlock Holmes solves the mystery if he notices small details and he catches the suspect. S2: Sherlock listens to Watson only if Watson has good ideas. S3: Watson will intervene if and only if Sherlock is in trouble or Watson has a new insight.
- [ ] **Set 4** — S1: Mary will start reading a fantasy book if and only if she finishes her current book and she has free time. S2: Jane will read the book if it has good reviews or is a bestseller. S3: Charlotte reads mystery books only if it is Sunday and she is bored.
- [ ] **Set 5** — S1: The wizard wears his starry hat only if he's outside and he is practicing magic. S2: The wizard's beard will sparkle if and only if he drinks enchanted tea or a unicorn sneezes nearby. S3: If the wizard uses the wrong ingredients or says the wrong words, then the spell will not work.
- [ ] **Set 6** — S1: If the team is losing or one of the players gets injured then the coach will change the strategy. S2: The team can make a comeback only if it improves its defense and scores more. S3: The team advances to the next round if and only if it wins the next game by 2 points or more.

Chapter 2 — Normal forms (CNF / DNF)

Exercise 6 — Prove validity via normal forms

Transform the formulas into their equivalent conjunctive and disjunctive normal forms. Using one of these forms prove that they are valid formulas.

- [ ] $U_1 = (p \to \neg q) \land (q \lor r) \to (p \to r)$
- [ ] $U_2 = (p \to q) \land (p \land q \to r) \to (p \to r)$
- [ ] $U_3 = (p \land q \to r) \to (p \to (q \to r))$
- [ ] $U_4 = (p \to (q \lor r)) \to ((p \to q) \lor (p \to r))$
- [ ] $U_7 = (p \to (q \land r)) \to ((p \to q) \land (p \to r))$ **(S)**
- [ ] $U_8 = p \lor (q \to r) \to ((p \lor q) \to (p \lor r))$

Exercise 7 — Models via the appropriate normal form

Using the appropriate normal form write all the models of the following formulas.

- [ ] $U_1 = (p \lor q \to r) \to (p \to r) \land q$
- [ ] $U_2 = \neg(\neg p \lor q) \lor r \to \neg p \land \neg(q \land r)$
- [ ] $U_3 = (p \land q \to r) \to (p \to r) \land q$
- [ ] $U_4 = (p \lor q) \land \neg r \to p \land q \land r$
- [ ] $U_5 = p \lor \neg(q \land \neg r) \to p \land q \land \neg r$
- [ ] $U_6 = (p \lor q \to r) \to (q \to r) \land p$

Exercise 8 — Prove inconsistency

Using the appropriate normal form, prove that the following formulas are inconsistent.

- [ ] $U_1 = (p \to (q \to r)) \land \neg((p \to q) \to (p \to r))$
- [ ] $U_3 = (p \to q) \land (p \land q \to r) \land (p \land \neg r)$
- [ ] $U_4 = (p \to (q \lor r)) \land (\neg(p \to q) \land \neg(p \to r))$
- [ ] $U_5 = p \land (q \to r) \land ((p \land q) \land \neg(p \land r))$
- [ ] $U_6 = (p \to (q \to r)) \land (p \land q \land \neg r)$
- [ ] $U_7 = (p \to (q \to r)) \land \neg(q \to (p \to r))$

Exercise 9 — Anti-models via the appropriate normal form

Write all the anti-models of the following formulas using the appropriate normal form.

- [ ] $U_1 = (q \land r \to p) \to (p \to r) \land q$
- [ ] $U_2 = (q \lor r \to p) \to (p \to r) \land q$
- [ ] $U_3 = (p \lor q \to r) \to (q \to r) \land p$
- [ ] $U_4 = p \lor \neg(q \land \neg r) \to p \land q \land \neg r$
- [ ] $U_5 = p \land \neg(q \land \neg r) \to p \lor q \land \neg r$
- [ ] $U_6 = (p \land q \to r) \to (p \to r) \land q$

Chapter 3 — Deduction (syntax)

Exercise 10 — Deductions by definition

Using the definition of deduction, prove the following deductions.

- [ ] $p \to q, r \to t, p \lor r, \neg q \vdash t$
- [ ] $p \to r, p \lor r \to q, r \vdash q$
- [ ] $q \to p, t \to r, q \lor t, \neg p \vdash r$
- [ ] $p \lor (q \to r), p \lor q, \neg p \vdash r$
- [ ] $\neg p \lor \neg q \lor r, q, p \vdash r$
- [ ] $p \to \neg q \lor r, p \land q, p \vdash r$

Exercise 11 — Is the conclusion deducible?

Is C deducible from the set of hypotheses {H1, H2, H3, H4}, i.e. H1, H2, H3, H4 $\vdash$ C? If yes, build its deduction.

- [ ] **Set 1** — H1: It is not sunny this afternoon and it is colder than yesterday. H2: We will go swimming only if it is sunny. H3: If we do not go swimming, then we will take a canoe trip. H4: If we take a canoe trip, then we will be home by sunset. C: We will be home by

sunset.

- [ ] **Set 2** — H1: Alice could have robbed the bank only if she left the house last night. H2: If Alice did not rob the bank, then the police will investigate the bank's employees. H3: In case the police investigate the bank's employees, they will question Bob. H4: Alice did not leave the house last night. C: The police will question Bob.
- [ ] **Set 3** — H1: If the dragon lives for many centuries, then he will accumulate a vast treasure. H2: The dragon can accumulate a vast treasure only if his lair remains undisturbed. H3: Provided that the neighbouring kingdoms are at war, the dragon's lair will not remain undisturbed. H4: The dragon has lived for many centuries. C: The neighbouring kingdoms are not at war.

Exercise 12 — Theorems via the deduction theorem

Prove the following theorems using the theorem of deduction and its reverse.

- [ ] $\vdash p \vee (q \to r) \to ((p \vee q) \to (p \vee r))$
- [ ] $\vdash (p \to (\neg r \to q)) \to (r \vee \neg p \vee q)$
- [ ] $\vdash (p \to (q \to r)) \to (p \wedge q \to r)$
- [ ] $\vdash (p \wedge q \to r) \to (p \to (q \to r))$
- [ ] $\vdash (p \to (q \to r)) \to (q \to (p \to r))$
- [ ] $\vdash (p \to (q \to r)) \to ((p \to q) \to (p \to r))$

Exercise 13 — More theorems via the deduction theorem

Using the theorem of deduction and its reverse prove that:

- [ ] $\vdash (p \to (q \vee r)) \to ((p \to q) \vee (p \to r))$
- [ ] $\vdash (p \to q) \to ((\neg r \vee p) \to (r \to q))$
- [ ] $\vdash p \vee (q \to r) \to ((p \vee q) \to (p \vee r))$
- [ ] $\vdash (p \to r) \to ((q \to r) \to (p \vee q \to r))$
- [ ] $\vdash (p \to q) \to ((r \to t) \to (p \wedge r \to q \wedge t))$
- [ ] $\vdash (p \to r) \to ((p \wedge r \to q) \to (p \to q))$

Part II — Predicate Logic

Chapter 4 — Formalization (natural language → predicate formulas)

Exercise 1 — Mathematical statements

Transform the sentences into predicate formulas. Explain the syntactic elements used: variables, constants, function symbols, predicate symbols.

- [ ] The sum and difference of two even numbers is an even number and their product is divisible by 4.
- [ ] If x is a nonzero integer divisible by 10, it can be decomposed in two factors such that one is divisible by 2 and the other one is divisible by 5, and x can be written as a sum of 2 even numbers.
- [ ] Every positive number can be written as a product of two positive numbers and as a product of two negative numbers.
- [ ] If x and y are positive prime numbers, $x \neq y$, $x \neq 2$ and $y \neq 2$, their sum and difference are even numbers and their product is an odd number.
- [ ] For every positive integer x, if x is a square of an integer, then there exists an integer y such that $(y + 1)\cdot(y - 1) = x - 1$. **(S)**
- [ ] If x is divisible by 3 and y is divisible by 6 then their product is an even number and their sum is divisible by 3.
- [ ] For every positive integer x, if x is not a prime, then there exists a prime y such that y divides x and $y < x$.

Exercise 2 — Everyday statements

Transform the statements into predicate formulas, choosing appropriate constants, function symbols and predicate symbols.

- [ ] Every student who makes good grades is brilliant or studies.
- [ ] Some of John's colleagues like to draw and some like to dance.
- [ ] CS students like either algebra or logic, all of them like Java but only Bill likes history.
- [ ] All Mary's relatives live in Cluj-Napoca, only her cousin John lives in Bucharest.
- [ ] Caterpillars and snails like to eat some plants.
- [ ] Anyone who owns a rabbit hates anything that chases a rabbit.
- [ ] Anyone who loves some candy is not a nutrition fanatic.
- [ ] If Santa has some reindeer with a red nose, then every child loves Santa.
- [ ] Anyone who has any cats will not have any mice.
- [ ] Caterpillars and snails are much smaller than birds, which are much smaller than foxes, which in turn are much smaller than wolves.
- [ ] Every animal either likes to eat all plants or all animals much smaller than itself that like to eat some plants.
- [ ] If every roommate of every CS major likes to party, then every student who is a CS major and makes good grades is brilliant.
- [ ] If Santa is happy at Christmas, then every child who is good gets some toy at Christmas.

- [ ] Wolves do not like to eat foxes or grains, while birds like to eat caterpillars but not snails.

Exercise 3 — Deduction in predicate logic

Check whether the conclusion C is derivable from the hypotheses using the definition of deduction and the appropriate inference rules.

- [ ] **Succession to the British throne.** H1: If x is the king and y is his oldest son, then y can become the king. H2: If x is the king and y defeats x, then y will become the king. H3: RichardIII is the king. H4: HenryVII defeated RichardIII. H5: HenryVIII is HenryVII's oldest son. C: Can HenryVIII become the king?

Chapter 5 — Semantics: interpretations, models, anti-models

Exercise 4 — Evaluate under a given interpretation

Using the given interpretations, evaluate the following formulas.

- [ ] **(a)** U = (∀x)(∀y)(A(x) ∧ A(y) → ¬A(sum(x, y)) ∧ (∀z)A(prod(y, z))); I = ⟨D, m⟩, D = ℕ; m(sum)(x, y) = x + y; m(prod)(x, y) = x·y; m(A)(x): "x is an odd number".
- [ ] **(b)** U = (∃x)(P(x) ∧ Q(x)) → (∃x)P(x) ∨ Q(12); D = ℕ; m(P)(x): "x is divisible by 5"; m(Q)(x): "x is divisible by 7".
- [ ] **(c)** U = (∃x)A(x) ∧ (∃x)B(x) → (∀x)(A(x) ∨ B(x)); D = the straight lines of a plane P; d ∈ D a fixed line; m(A)(x): "x ⊥ d"; m(B)(x): "x ∥ d".
- [ ] **(d)** U = (∀x)(P(x) ∧ Q(x) → P(sq(x)) ∧ Q(prod(x, 5))); D = ℤ; m(P)(x): "x is even"; m(Q)(x): "x < 0"; m(prod)(x, y) = x·y; m(sq)(x) = x². **(S)**
- [ ] **(e)** U = (∀x)(A(x) → B(x)) ∧ (∃x)(B(x) ∧ ¬A(x)); D = ℕ; m(A)(x): "x is a prime"; m(B)(x): "x is an odd number". **(S)**
- [ ] **(f)** U = ((∀x)A(x) → (∃x)B(x)) → ((∃x)A(x) → (∀x)B(x)); D = all persons from Romania; m(A)(x): "x likes chess"; m(B)(x): "x is a student".
- [ ] **(g)** U = (∃x)A(x) ∧ (∃x)B(x) → (∀x)(A(x) ∧ B(x)); D = all persons from Romania; m(A)(x): "x owns a car"; m(B)(x): "x has a driver licence".

Exercise 5 — A model on a 2-element domain

Choose an arbitrary interpretation with a finite domain (2 elements) for the formula U_j and prove that it is a model of U_j.

- [ ] U_1 = (∀x)(A(x) ↔ B(x)) → ((∀x)A(x) ↔ (∀x)B(x))
- [ ] U_2 = (∀x)(A(x) → B(x)) → ((∀x)A(x) → (∀x)B(x))
- [ ] U_3 = (∀x)(A(x) ↔ B(x)) → ((∃x)A(x) ↔ (∃x)B(x))
- [ ] U_4 = (∃x)(A(x) → B(x)) ↔ ((∀x)A(x) → (∃x)B(x))

- [ ] $U_5 = ((\exists x)A(x) \to (\forall x)B(x)) \to (\forall x)(A(x) \to B(x))$ **(S)**
- [ ] $U_6 = (\forall x)(A(x) \vee B(x)) \to ((\forall x)A(x) \vee (\exists x)B(x))$
- [ ] $U_7 = (\forall x)(A(x) \to B(x)) \to ((\exists x)A(x) \to (\exists x)B(x))$
- [ ] $U_8 = (\forall x)(A(x) \to B(x)) \to ((\forall x)A(x) \to (\exists x)B(x))$

Exercise 6 — A model and an anti-model

Find a model and an anti-model for the following formulas.

- [ ] $U_1 = ((\exists x)P(x) \to (\exists x)Q(x)) \to (\forall x)(P(x) \to Q(x))$
- [ ] $U_2 = (\exists x)(P(x) \to Q(x)) \to ((\exists x)P(x) \to (\exists x)Q(x))$
- [ ] $U_3 = ((\forall x)P(x) \to (\forall x)Q(x)) \to (\forall x)(P(x) \to Q(x))$
- [ ] $U_4 = (\exists x)P(x) \wedge (\exists x)Q(x) \to (\exists x)(P(x) \wedge Q(x))$
- [ ] $U_5 = (\forall x)(P(x) \vee Q(x)) \to (\forall x)P(x) \vee (\forall x)Q(x)$ **(S)**
- [ ] $U_6 = ((\forall x)P(x) \to (\exists x)Q(x)) \to (\forall x)(P(x) \to Q(x))$
- [ ] $U_7 = (\exists x)P(x) \wedge (\exists x)Q(x) \to (\forall x)(P(x) \wedge Q(x))$

Extra — Seminar exercises not on the sheets

Chapter 6 — Semantic tableaux

- [ ] **(S)** Using the semantic tableaux, decide what kind of formula is $U = (r \vee q) \vee (p \to \neg r) \to (p \leftrightarrow q)$ (consistent, inconsistent, valid). If U is consistent, find its models.
- [ ] Re-do Exercise 1 (U_1–U_6) and Exercise 8 with semantic tableaux, and check that you get the same answers as with truth tables / normal forms.